Analysis of current harmonics in data center power system

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***Abstract* - This paper presents power quality analysis at different points in a data center. Electrical distribution network scheme and data center description are also given. The method of measurements performed is briefly described and the results of real measurements of current and voltage Total Harmonic Distortion (THD) are presented and compared with IEEE 519-2014 recommended practice and requirements for harmonic control in electric power system. All measured values comply with the standard but measurements at downstream levels closer to the nonlinear loads have larger THD.**

Keywords - data center; power quality; current harmonic distortion; nonlinear loads

I. INTRODUCTION

With the development of digital technology, data centers are becoming increasingly important. Most of the world's Internet Protocol (IP) traffic goes through data centers. For every bit of data that travels the network, another five bits of data are transmitted within and among data centers. Greater connectivity is propelling demand for data center services and energy consumption (primarily electricity). In the last decade, global Internet traffic has increased 15-fold, and only in 2020 it increased by more than 40%, mainly because of social networking and video streaming growth. Globally, data transmission networks consumed 260-340 TWh in 2020, equivalent to 1.1-1.4% of global electricity consumption [1]. Data transmission network technologies are also becoming more efficient. Total Harmonic Distortion (THD) is closely related to efficiency and power quality. Harmonics come in two basic varieties: current distortion and voltage distortion. Nonlinear loads in data centers, servers, and IT equipment, are the most common source of current distortion and additional losses in the data center power system. In this paper, special attention has been paid to current distortion.

Data center power quality is a widespread concern. The power quality of Libyan Telecom Technology headquarter data center is analysed [2]. The total current and voltage harmonic distortions were measured at points with different loads and compared with the recommended values. An isolation transformer for harmonic separation and series passive filters for harmonic reduction were recommended. In another data center, the power quality of IT equipment with and without Active Power Factor Control (APFC) is analysed [3]. A reactive power compensator was indicated to achieve reactive power compensation and reduce the predominant harmonic level. The cooling system is another type of nonlinear load [4]. A passive harmonic filter designed specifically for the low DC bus capacitance of Electronically commutated (EC) fans can reduce the cooling system negative effects. Factory testing confirmed the accuracy of the computer simulations and the IEEE Std 519 recommendations have been met. In addition to passive filters, Active Harmonic Filters (AHFs) and hybrid filters are also used to reduce distortions [5]. AHF's with harmonic cancelation transformers proved to be very efficient in reducing harmonics in data center. It was confirmed in 2007 in a medium legacy data center where the THD was increased by nonlinear loads outside the center.

Variable frequency drives are used in data centers to improve the efficiency of cooling and ventilation systems [6]. But these drives also cause harmonics. THD can be reduced with active and passive filters. An active filter is an effective way to eliminate both harmonics and reactive currents from the AC source, and it can be applied to the entire power system. For passive filters in combination with harmonic cancelation transformers, special attention should be paid to load balancing. Hybrid Power Filter (HPF) is a combination of Active Power Filter (APF) and Passive Power Filter (PPF), which is mainly used to reduce 3rd and 5th harmonic and improve power quality [7]. 3rd harmonic caused heating of neutral conductors and transformers in a data center. HPF reduces current in neutral and is preferable in terms of low cost, flexibility, reliability and maintenance.

Data center power quality analysis is presented in the paper. It has been shown that the THD value is lower than the recommended values at all levels of the system, but significantly higher at downstream levels than at upstream levels. Upper THD limits are recommended in the literature [5], [8]. The paper is divided into five sections. Section I consists of an introduction and related work. In Section II, part of the electrical distribution network scheme is presented. Section III presents the analysis of

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power quality at an upstream level of the system. The Uninterruptible Power Supply (UPS) room power quality analysis is given in the Section IV, while Section V contains the power quality analysis at the downstream level of the system. Future work and conclusion are given in Section VI.

II. DATA CENTER DESCRIPTION

Figure 1 shows one of two equally independent data center branches, branch "A". This provides redundancy in powering the entire system. Each branch has its own auxiliary generator to provide power to the system in the event of a power failure. The branches have three UPSs, two of which are currently in operation. UPSs power both, the IT equipment and the air conditioning units to maintain the temperature in the UPS room. The measurement points are also marked in the figure. The measurements are divided into levels, where "distribution room" is the highest level and "server room" is the lowest level of the system. P1 point presents the Point of Common Coupling (PCC).

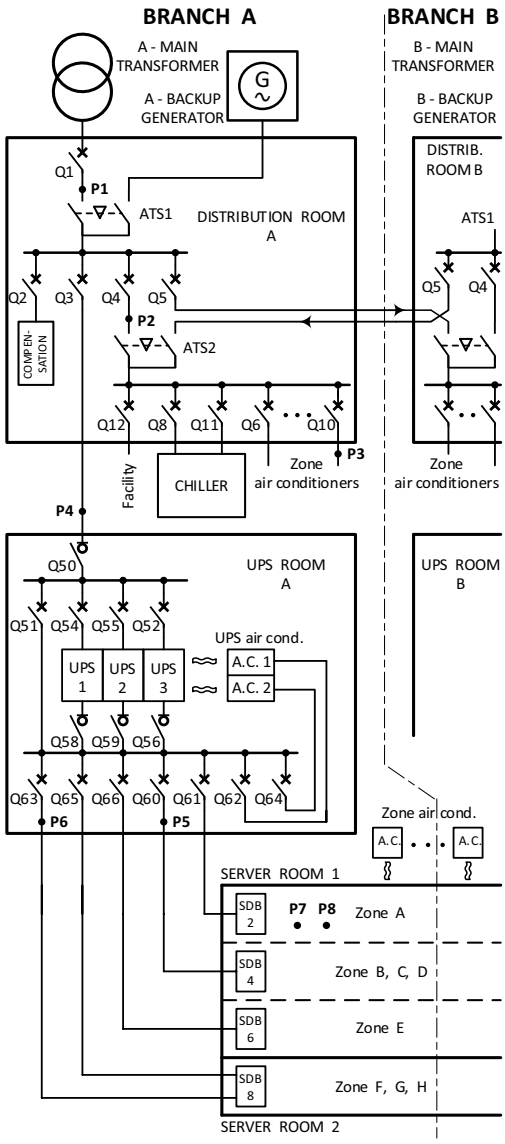


Figure 1. Branch "A" - electrical distribution network

Reference measurements were made within one week using a 10-minute average. Measurements were made using a Chauvin Arnoux C.A 8336 network analyzer. Given the typical consumption, measurements were repeated at the time of peak consumption to obtain a picture of harmonic distortion. Repeated measurements were made at two-minute intervals using Dewesoft XHS (HIGH-SPEED DATA ACQUISITION SYSTEM) and a sampling frequency of 20kHz.

III. DISTRIBUTION ROOM (HIGHEST LEVEL) MEASUREMENTS

A. *Measurement point P1*

The load is about 20% of the transformer power. The difference in the load of each phase is only a few percent and has no technical significance. Asymmetry of the load is inevitable due to a certain number of single-phase loads. In terms of total installed power, the largest part consists of three-phase balanced loads (UPS and air conditioners). The current THD is 4.2-4.5% (TABLE 1). The most pronounced current harmonic is H5 (Figure 2); stepwise it ranges from 1.5% to 2.8%. It is followed by H13 (2%), H11 (about 1.5%) and all others are less than 1% (except H7 in phase 2). The most pronounced voltage harmonic is H7 with values of 1.5-1.8%. All other harmonic values are less than 0.3%. The current and voltage THD values measured at the Point of Common Coupling (PCC) are in accordance with the IEEE std. 519-2014 for General Distribution Systems up to 69kV. The network short circuit power is much higher than the power consumption and therefore the power quality cannot be affected.

B. *Measurement point P2*

Switch Q4 supplies the bus to which all cooling units are connected. Therefore, the distribution of harmonics is the same as in "*P1*", but H3 in phase 2 is slightly smaller (Figure 3). There is also a frequency component of 6 kHz (Figure 5).

C. *Measurement point P3*

The load at this circuit is an air conditioner for a zone. From the measured data (power, current, THD), it appears that it is a combination of three-phase balanced load and single-phase load in phase 2. The influence of this load is most evident at H3 and H9 (Figure 4). In phases 1 and 3, the magnitudes of the individual harmonics are the same. The increased component of the 3rd harmonic in phase 2 can be seen in the current waveform. There is also a 6 kHz component in the current generated by chopper assemblies inside the air conditioner (Figure 6). Voltage total harmonic distortion (THDv) values for all three points at this level for phases 1, 2 and 3 are 1,5%, 1,7% and 1,6%.

D. *Measurement point P4*

The UPS bus is powered by a switch Q50. The load is balanced, practically fully resistive (the current lags the voltage by only 0.6°), and the current THD is 3.5-4.2% (TABLE 2). During the measurement, this load accounted for 70-80% of the total transformer load. Proportion of

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balanced load with low harmonic level has an extremely favorable effect on the quality of power supply from DC! Looking at the harmonics, H7 is the largest (over 3%), H13 is around 2%, H5 and H11 are between 1% and 2%, while the rest are well below 1% (Figure 7).

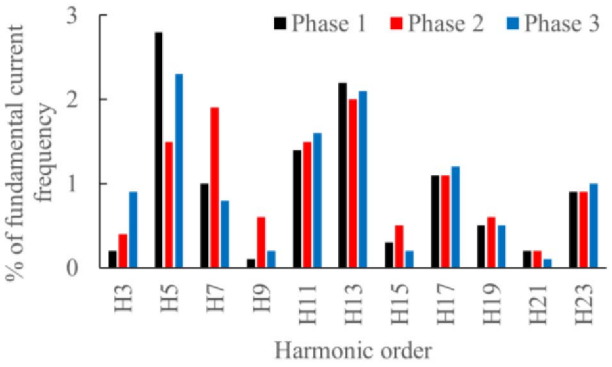


Figure 2. Harmonic order for *P1*

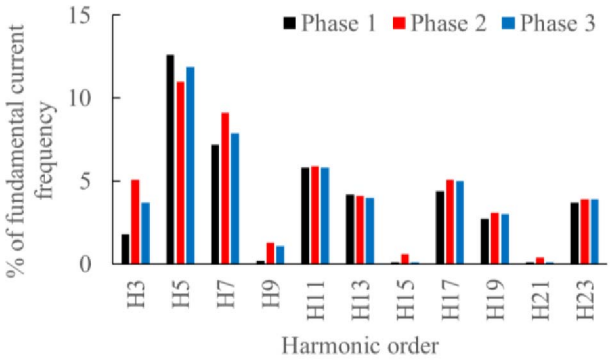


Figure 3. Harmonic order for *P2*

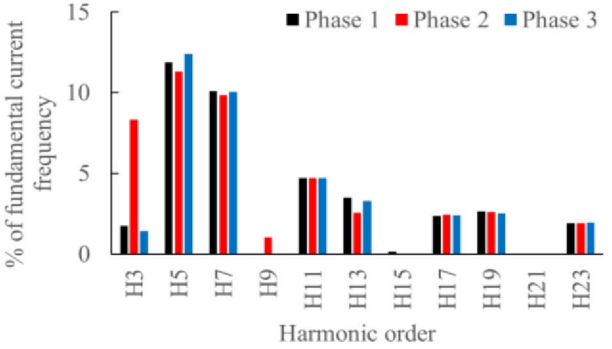


Figure 4. Harmonic order for *P3*

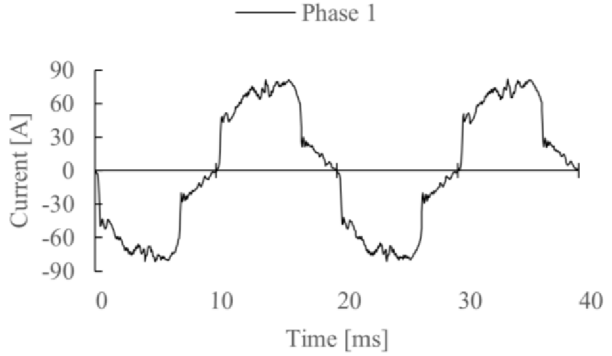


Figure 5. Current waveform for *P2*

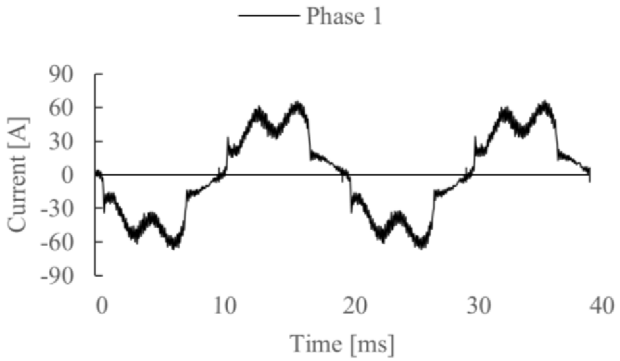


Figure 6. Current waveform for *P3*

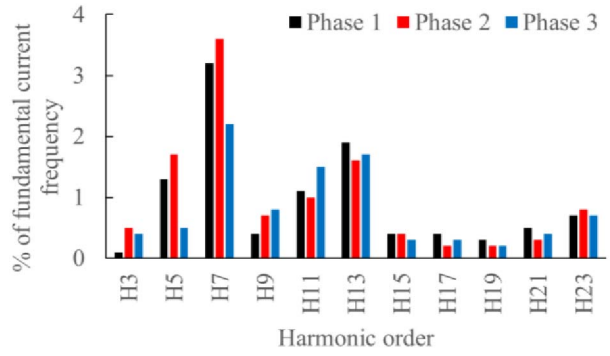


Figure 7. Harmonic order for *P4*

TABLE 1. Distribution room measured values

P	S [kVA]	P [kW]	Q [kVAr]	THDi [%] L1	L2	L3
1	189	188,4	11,3	4,4	4,2	4,5
2	40,2	38,9	10,2	19,8	20,8	20,1
3	28	25,6	11,2	24	24,5	23,5
4	136,3	136,2	1,5	3,98	4,2	3,52

IV. UPS ROOM (MIDDLE LEVEL) MEASUREMENTS

A. Measurement point P5

By the Q60 switch, the UPS output voltage is transmitted to the zone busbar SDB4 in the server room. All loads connected to SDB4 are single phase (power supplies for servers and IT equipment), but the distribution by phase is such that the phase loads differ by less than 10% (Figure 8). The current distortion is 8.6-9.5%. H3 and H11 are dominant in the spectrum (Figure 9).

B. Measurement point P6

Switch Q63 supplies the circuit from UPS to the SDB8 (zones F, G and H). The loads (servers power supplies) connected to SDB8 are newer generations with lower current THD. The load of phase 3 is the highest, and the loads of phases 1 and 2 are about half the value of the phase 3 load (Figure. 10), so the THD for phase 3 is the lowest at 4.9%, while it is 10% and 7% for phases 1 and 2 (Figure 11). These values are better than those obtained by measurement at the previous point. Voltage

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total harmonic distortion (THDv) values for all three points at this level for phases 1, 2 and 3 are 1,78%, 1,74% and 1,6%.

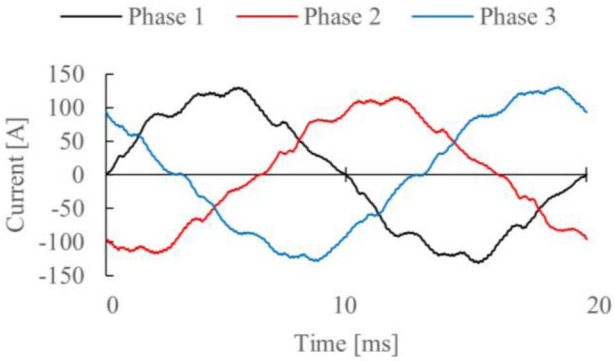


Figure 8. Current waveform for *P5*

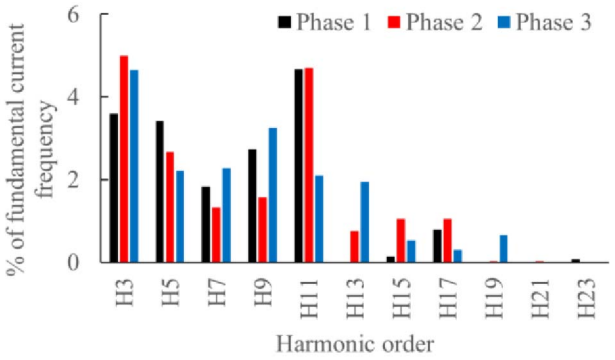


Figure 9. Harmonic order for *P5*

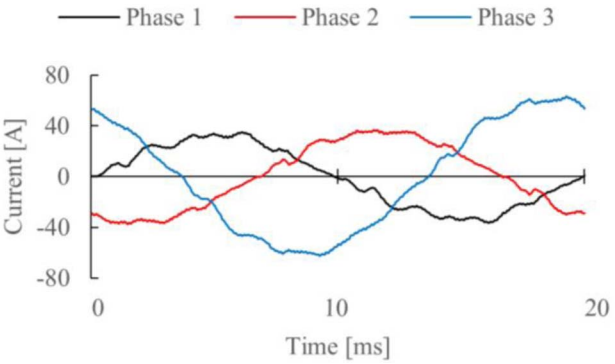


Figure. 10 Current waveform for *P6*

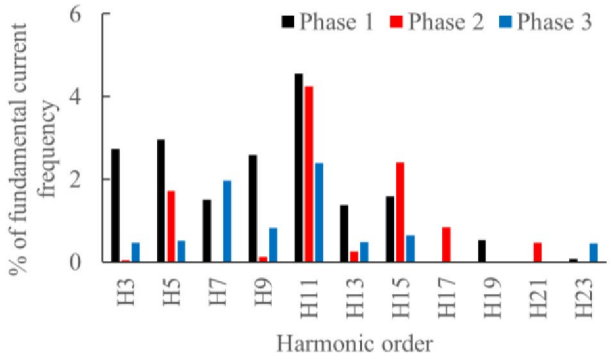


Figure 11. Harmonic order for *P6*

TABLE 2. UPS room measured values

P	S [kVA]	P [kW]	Q [kVAr]	THDi [%] L1	L2	L3
5	56,3	53,6	16,9	9,3	9,5	8,6
6	21,1	20,3	5,7	10	7	4,9

V. SERVER ROOM (LOWEST LEVEL) MEASUREMENTS

A. Measurement point P7

This measurement point is located on a single-phase circuit supplying multiple supplies in a rack. The reactive power is capacitive, and its magnitude is approximately equal to the active power. The current leads the voltage by 47°. Figure 12 shows the current waveform. The small amount of active power and the large current displacement are typical for lightly loaded supplies. The THD of the current is 46% (

). The harmonic H3 is about 25% and H5, H7 and H11 about 14% (Figure 13). The voltage THD is 2.14%, which is higher than UPS (Q60) due to additional voltage drops caused by current harmonics.

B. Measurement point P8

The last measurement point is on a single-phase circuit for powering multiple newer generation power supplies in a rack. The waveform of the current is nearly sinusoidal (Figure 14). The supplies are moderately loaded. The current leads the voltage by about 15°. The individual harmonics are less than 5% (Figure 15) and the total harmonic distortion is less than 10%. Voltage total harmonic distortion (THDv) value is 2,04% for both measurement points.

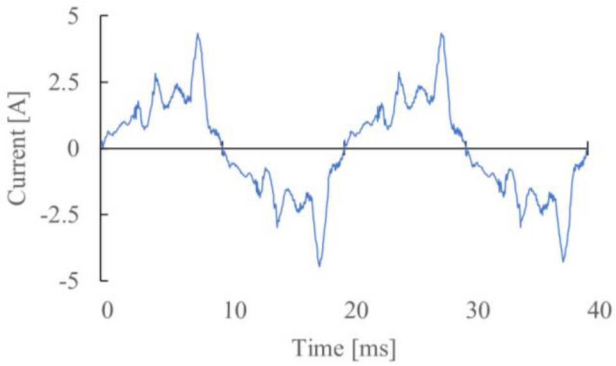


Figure 12. Current waveform for *P7*

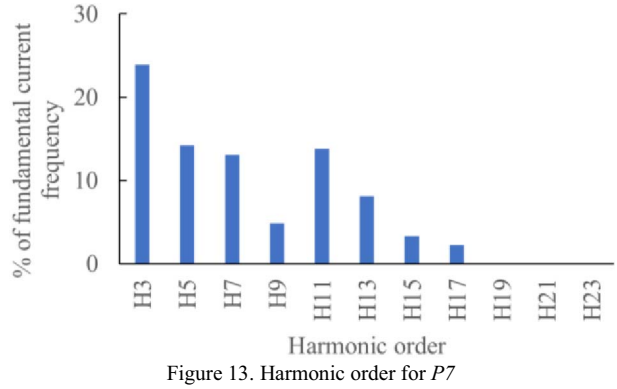


Figure 13. Harmonic order for *P7*

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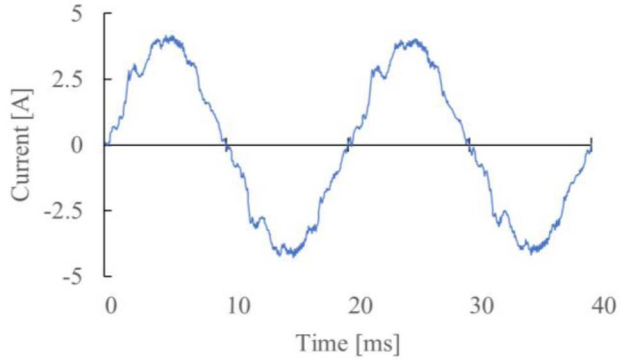


Figure 14. Current waveform for *P8*

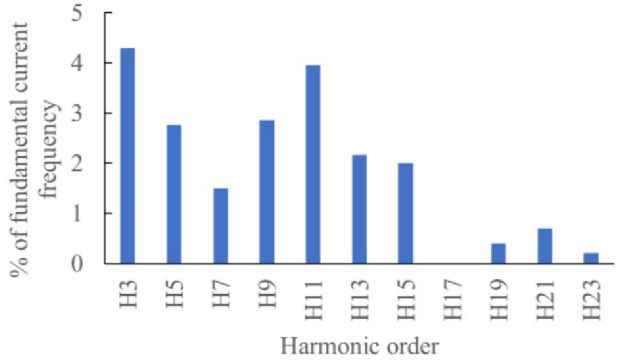


Figure 15. Harmonic order for *P8*

TABLE 3. Server room measured values

P	S [kVA]	P [kW]	Q [kVAr]	THDi [%]
7	0,42	0,29	0,31	46
8	0,62	0,6	0,17	9,5

VI. CONCLUSION

In this paper, power quality analysis at different points in a data center is given. The current and voltage THD values measured at the Point of Common Coupling (PCC) are in accordance with the IEEE std. 519-2014 for General Distribution Systems up to 69kV. The busbar harmonic distortion upstream of the Uninterruptible Power Supplies (UPS) is low, since each UPS has its own internal Power Factor Correction (PFC). However, measurements show that points at downstream levels closer to nonlinear loads such as servers or air conditioners, have larger current harmonic distortion values and contribute to an increase in system energy losses. Future work will be to develop a filter for harmonic reduction and test its functionality in the analyzed data center.

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